# AIDAN SCANNELL

**Machine Learning Researcher**

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*"Machine Learning Researcher with 6+ years of experience developing and implementing machine learning and sequential decision-making algorithms, with a focus on learning world models and leveraging them for downstream decision-making. Proven ability to translate research ideas into high-performing implementations on multi-GPU systems, including generative modelling approaches such as Transformers. Experienced in leading research teams, driving collaborations, and presenting work at top-tier conferences."*

## SKILLS

Python · C++ · PyTorch · JAX · Hydra · Weights & Biases · Docker · SLURM · Git · LaTeX · ROS

Distributed ML · AMP · World models · Reinforcement learning · Generative modelling · Embodied AI

## EXPERIENCE

### Research Associate with Prof. Amos Storkey & Prof. Peter Bell

**University of Edinburgh | Bayesian and Neural Systems group**

Jan 2025 – Ongoing · Edinburgh, UK

- Won the 1X World Model Challenge and presented at ICCV 2025 workshop "Learning to See: Advancing Spatial Understanding for Embodied Intelligence".
- Developed Transformer- and diffusion-based world models for long-horizon and controllable generation.
- Bridged algorithmic design and scalable implementation, achieving 3× throughput in multi-GPU training pipelines.
- Presented invited talks on world models and scalable generative training at academic and industry venues.

World models · Transformers · Generative modelling · Scalable training · Multi-GPU systems · Embodied AI

### Postdoctoral Researcher with Prof. Joni Pajarinen & Prof. Arno Solin

**Aalto University | Finnish Center for Artificial Intelligence (FCAI)**

July 2022 – December 2024 · Helsinki, Finland

Received four-year funding to work jointly in the Robot Learning Lab and Machine Learning Group.

- Led cross-disciplinary collaborations with multiple PIs and PhD students to develop a generative world model for both decision-time and background planning; published at ICLR 2025.
- Co-led research on Bayesian deep learning for uncertainty-aware world models to guide exploration in model-based RL; published at ICLR 2024.
- Supervised MSc and PhD research on robotics foundation models, RL, and generative modeling for decision-making.
- Led FCAI's "Long-term Decision Making and Transfer Between Tasks" research team, defining technical strategy and mentoring early-career researchers.
- Scaled and optimized training on multi-GPU HPC clusters using PyTorch for world model and RL experiments.
- Maintained research codebases with documentation (Sphinx), testing, and open-source collaboration.

World models · Generative modelling · Reinforcement learning · Scalable training · Robotics · Leadership

### Co-lecturer @ **Aalto University**

Sept 2022 – December 2024 · Helsinki, Finland

- Co-lecturer for courses on *Reinforcement Learning* and *Advanced Gaussian Processes*, covering both theoretical foundations and practical implementation.
- Delivered lectures and tutorials, contributing to course design and receiving strong student feedback.

Teaching · Communication · Mentorship · Public speaking · Reinforcement learning

### PhD Researcher (Supervisors: Prof. Arthur Richards & Prof. Carl Henrik Ek)

**CDT in Future Autonomous and Robotic Systems, University of Bristol/Bristol Robotics Laboratory**

Sept 2018 – May 2022 | Bristol, UK

Awarded a four-year PhD scholarship including a taught MRes year.

- Combined methods from machine learning, stochastic geometry, and reinforcement learning to control quadcopters in uncertain real-world environments.
- Deployed learning-based control algorithms on physical robots using TensorFlow, JAX, and ROS.
- Presented findings at AISTATS and ICRA, bridging ML theory and robotic control.
- Teaching assistant for *Machine Learning*, *Robotic Systems*, and *Intelligent Information Systems* courses.

Machine learning | Reinforcement learning | Uncertainty quantification | Robotics | Stochastic geometry

## AWARDS

- Won both tracks of the 1X World Model Challenge and presented findings at ICCV 2025 Workshop.
- Awarded >300k GPU hours by LUMI Supercomputer for “”Sample-efficient Large-scale Reinforcement Learning”.
- Awarded Postdoctoral Networking Tour in Artificial Intelligence (Postdoc-NeT-AI) fellowship by the German Academic Exchange Service (DAAD) for funded research visit to Germany.
- Received 4-year postdoctoral funding from Finnish Center of AI.
- Awarded a four-year PhD scholarship at University of Bristol with the FARSCOPE Center for Doctoral Training.

## INVITED TALKS

- Generative World Models, Huawei AI Application Workshop, 26th November 2025, Dublin, Ireland
- Generative World Modelling for Humanoids, ICCV 2025 Workshop on Learning to See: Advancing Spatial Understanding for Embodied Intelligence, 19th October 2025, Hawaii, USA
- Discrete Codebook World Models, Huawei-University of Edinburgh Joint Lab Workshop 2025, 2nd April 2025, Xi’an, China
- Sample-Efficient RL with Implicitly Quantized Representations, Nordic AI Meet + AI Day 2024, 20th Oct 2024, Helsinki, Finland
- iQRL - Implicitly Quantized Representations for Sample-efficient RL, International Workshop of Intelligent Autonomous Learning Systems, 23 July 2024, Darmstädter Haus, Austria
- Model-based RL, Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2024, 17th July 2024
- Sequential Decision Making - Bayesian Optimization & Model-based RL, Advanced course on Gaussian processes @ Aalto University, 8 April 2024
- (Function-space) Laplace Approximation for BNNs, National Science Foundation (NSF) Safe RL Team, 3 Oct 2023
- Neural Networks as Sparse Gaussian processes for Sequential Learning, International Workshop of Intelligent Autonomous Learning Systems, 15 Aug 2023, Darmstädter Haus, Austria
- Model-based RL under uncertainty: the importance of knowing what you don’t know, RL course @ Aalto University, Nov. 2023
- Sequential Decision Making, Advanced course on Gaussian processes @ Aalto University, 4 April 2023
- Model based RL under uncertainty, ML at the Cambridge Computer Lab (ML@CL), 23 Feb 2023, University of Cambridge
- Synergising Bayesian Inference and Probabilistic Geometries for Robotic Control, Cognitive Systems Group @ Technical University of Denmark (DTU), 18 March 2021

## EDUCATION

### PhD in Robotics and Autonomous Systems

**University of Bristol**

Sept 2018 - June 2022

**PhD Thesis**:
- Bayesian Learning for Control in Multi-modal Dynamical Systems

**Taught MRes Year**:
- First class honours (GPA=4.0/4.0)

**Summer Schools**:
- Machine Learning Summer School 2019
- Gaussian Process and Uncertainty Quantification Summer School 2019

### MEng in Mechanical Engineering

**University of Bristol | First Class Honours (GPA=4.0/4.0)**

Sept 2012 – June 2016

- Graduated in top 10% of cohort

## REFERENCES

**Prof. Arno Solin**
Aalto University
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**Prof. Joni Pajarinen**
Aalto University
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**Prof. Carl Henrik Ek**
University of Cambridge
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**Prof. Arthur Richards**
University of Bristol
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# REVIEWING

- Area chair for European Workshop on Reinforcement Learning (EWRL)
- International Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- (Senior Reviewer) Reinforcement Learning Conference (RLC)
- IEEE Transaction on Pattern Analysis and Machine Intelligence (TPAMI)
- Annual Conference on Learning for Dynamics and Control (L4DC)
- International Conference on Artificial Intelligence and Statistics (AISTATS)
- Conference on Robot Learning (CoRL)
- International Conference on Robotics and Automation (ICRA)
- IEEE Robotics and Automation Letters (RA-L)

# PUBLICATIONS

## Publications


- Riccardo Mereu*, **Aidan Scannell***, Yuxin Hou, Yi Zhao, Aditya Jitta, Antonio Dominguez, Luigi Acerbi, Amos Storkey, and Pauland Chang (2025). Generative World Modelling for Humanoids: 1X World Model Challenge Technical Report. In: *arxiv preprint arXiv:2510.07092*. * Equal contribution.
- Mohammadreza Nakhaeinezhadfard, **Aidan Scannell**, and Joni Pajarinen (2025a). Contextual Latent World Models for Offline Meta Reinforcement Learning. In: *under review*.
- – (2025b). Offline Meta-Reinforcement Learning in Structured Non-Stationary Environments. In: *under review*.
- Ben Sanati, Thomas Lee, Trevor McInroe, **Aidan Scannell**, Nikolay Malkin, David Abel, and Amos Storkey (2025). Forgetting is Everywhere. In: *under review*.
- **Aidan Scannell**, Mohammadreza Nakhaei, Kalle Kujanpää, Yi Zhao, Kevin Luck, Arno Solin, and Joni Pajarinen (2025). Discrete Codebook World Models for Continuous Control. In: *The Thirteenth International Conference on Learning Representations*.
- Yi Zhao, **Aidan Scannell**, Yuxin Hou, Tianyu Cui, Le Chen, Arno Solin, Juho Kannala, and Joni Pajarinen (2025). Generalist World Model Pre-Training for Efficient Reinforcement Learning. In: *ICLR 2025 Workshop on World Models: Understanding, Modelling and Scaling*.
- Yi Zhao, **Aidan Scannell**, Wenshuai Zhao, Yuxin Hou, Tianyu Cui, Le Chen, Arno Solin, Juho Kannala, and Joni Pajarinen (2025). Efficient Reinforcement Learning by Guiding Generalist World Models with Non-Curated Data. In: *arXiv preprint arXiv:2502.19544*.
- Mohammadreza Nakhaei, **Aidan Scannell**, and Joni Pajarinen (2024a). Entropy Regularized Task Representation Learning for Offline Meta-Reinforcement Learning. In: *AAAI Conference on Artificial Intelligence (AAAI)*.
- – (June 2024b). Residual Learning and Context Encoding for Adaptive Offline-to-Online Reinforcement Learning. In: *Proceedings of The 6th Annual Conference on Learning for Dynamics and Control*.
- **Aidan Scannell**, Kalle Kujanpää, Yi Zhao, Mohammadreza Nakhaei, Arno Solin, and Joni Pajarinen (2024a). iQRL - Implicitly Quantized Representations for Sample-efficient Reinforcement Learning. In: *arXiv preprint arXiv:2406.02696*.
- – (July 2024b). Quantized Representations Prevent Dimensional Collapse in Self-predictive RL. in: *ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET 2024)*.
- **Aidan Scannell**, Riccardo Mereu, Paul Chang, Ella Tami, Joni Pajarinen, and Arno Solin (May 2024). Function-space Parameterization of Neural Networks for Sequential Learning. In: *The Twelfth International Conference on Learning Representations (ICLR)*.
- **Aidan Scannell**, Carl Henrik Ek, and Arthur Richards (Apr. 2023). Mode-constrained Model-based Reinforcement Learning via Gaussian Processes. In: *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- **Aidan Scannell**, Riccardo Mereu, Paul Chang, Ella Tami, Joni Pajarinen, and Arno Solin (July 2023). Sparse Function-space Representation of Neural Networks. In: *ICML 2023 Workshop on Duality Principles for Modern Machine Learning*.
- **Aidan Scannell** (2022). "Bayesian Learning for Control in Multimodal Dynamical Systems". PhD thesis. University of Bristol.
- **Aidan Scannell**, Carl Henrik Ek, and Arthur Richards (June 2021). Trajectory Optimisation in Learned Multimodal Dynamical Systems Via Latent-ODE Collocation. In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*.